NL RESILIENCE LAB

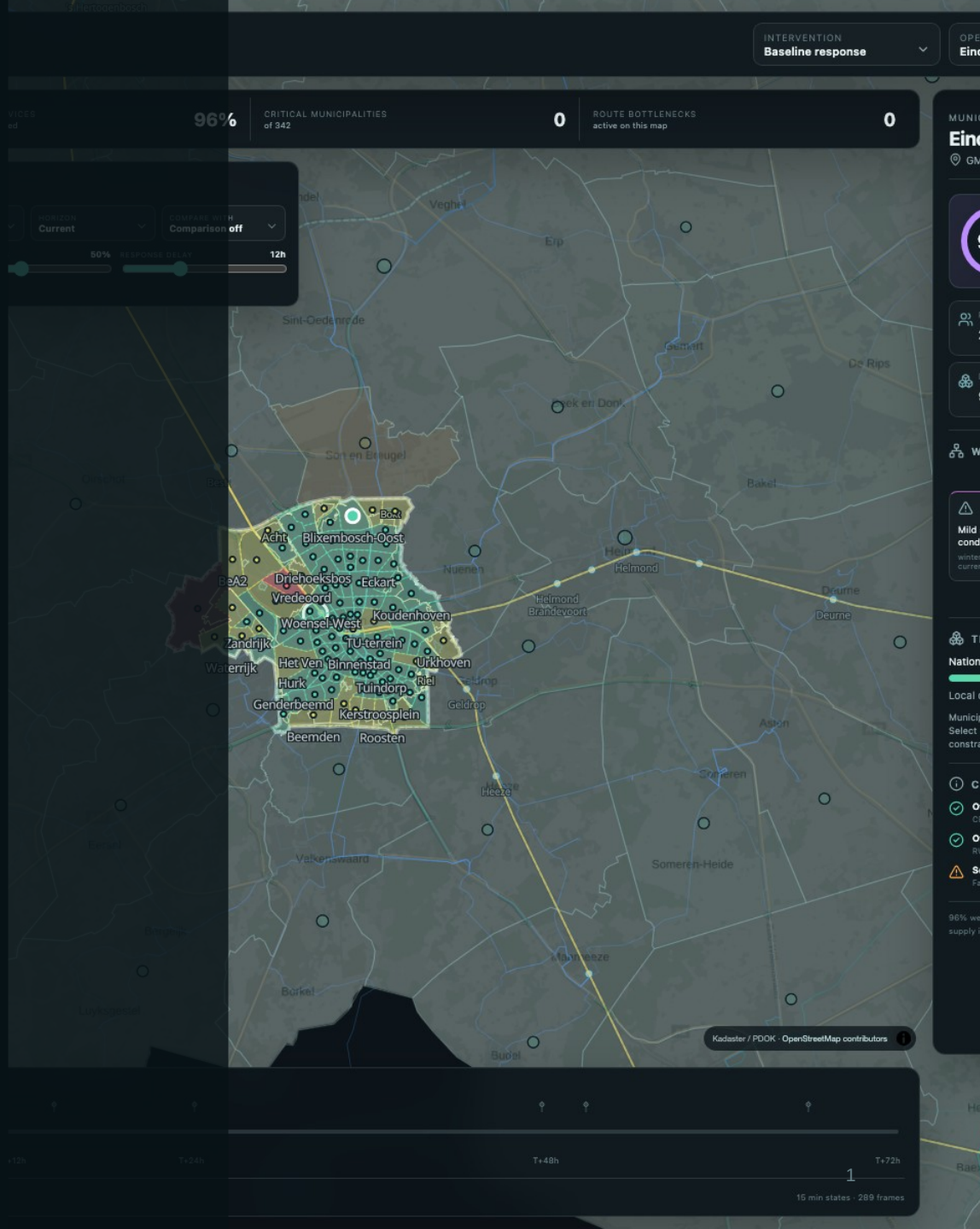
Exploring the first 72 hours

A transparent disaster-cascade simulator for the Netherlands

- SCENARIO EXPLORATION
- EXPLAINABLE MODEL
- OFFICIAL DUTCH DATA

Rik Stokmans / DisasterSimulator

github.com/Rik-Stokmans/DisasterSimulator



WHY

A disaster is not one failure.

It is a sequence of dependencies, delays and constrained choices.

01

INITIATING EVENT

Wind, water, heat, cyber, ground motion or fire

→

02

SERVICE CASCADE

Power, telecom, water, mobility and supply degrade on different clocks

→

03

HUMAN PRESSURE

Consequences emerge from duration, geography and response—not severity alone

CORE QUESTION **How does the system evolve—and which intervention changes the outcome?**

OVERVIEW

The simulator at a glance

A national model that stays inspectable down to municipality and CBS neighbourhood scale.

342
MUNICIPALITIES

72 h
SIMULATION WINDOW

289
15-MINUTE STATES

10
SCENARIO PRESETS

8
ESSENTIAL SERVICES

11
MAP QUESTIONS



Counts from the current repository snapshot · 4 September 2026

GEOGRAPHY

One model, three spatial scales

The same time, service state and intervention remain coherent while the view changes.

01

NETHERLANDS

342 municipalities
National service balancing
Affected-region footprint

02

MUNICIPALITY

One stable gateway
Demand, delivery and reserve
Local causal trace

03

CBS BUURT

Population + vulnerability weights
Local impact allocation
Official regional context

CONSERVATION RULE **Local allocation redistributes municipal inflow; it never creates supply.**

ENGINE

How a simulation is produced

Each quarter-hour frame is a deterministic sequence of visible model stages.



01 SCENARIO

Hazard, season, climate, intensity, start and duration

02 HAZARD

Time envelope + spatial eligibility + audited guards

03 DAMAGE

Service-specific pressure × resilience and preparedness

04 CASCADE

Power and mobility propagate into dependent services

05 BALANCE

Conservative route transfers reduce local shortfalls

06 EXPLAIN

Impacts, status, causes and counterfactuals

REPRODUCIBILITY

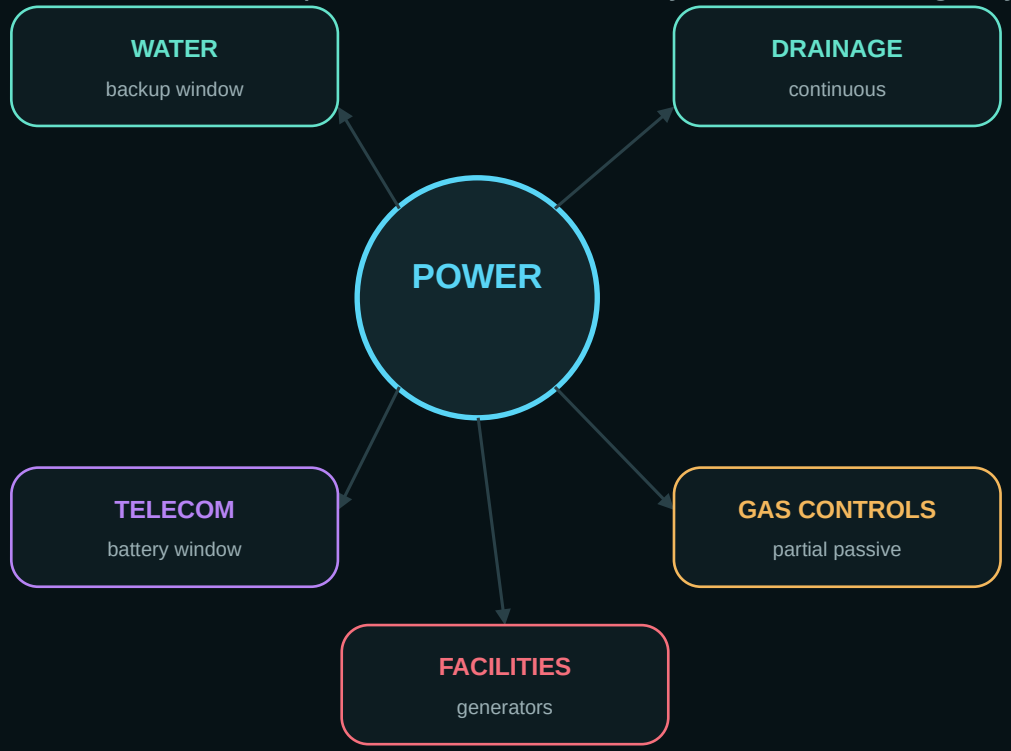
same configuration + same stable seed → same state and same explanation

Deterministic means repeatable—not calibrated or forecast-certain.

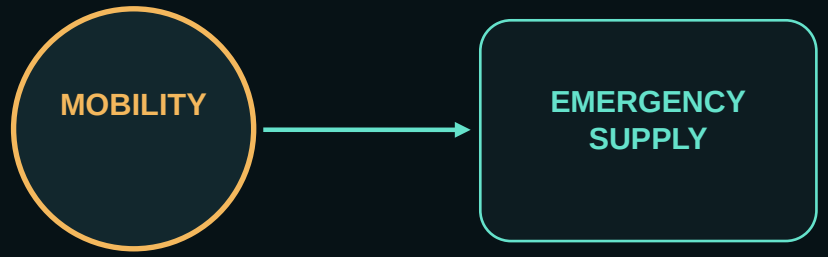
DEPENDENCIES

The cascade has two structural hubs

Power constrains dependent services; mobility constrains emergency supply.



DIRECT HAZARD PRESSURE CAN AFFECT EVERY SERVICE



Road access + logistics

NWB corridors are causal only for Mobility and explicitly modeled road logistics / Supply.

TIME

The answer changes across 72 hours

Shock, backup endurance, cascading shortfall and restoration happen on different clocks.

289 quarter-hour states



RESPONSE DELAY

sets when organized restoration may begin

BACKUP ENDURANCE

temporarily decouples dependent services from power

ACCESS GUARD

standing-water stress can pause generic repair

PEAK DAMAGE

can persist after current forcing has declined

COUPLING RULES

What actually drives the model?

The simulator separates causal inputs from visible evidence and from unresolved gaps.

affectsSimulation = false by default for operational context

CAUSAL	**Scenario + model inputs**	Disaster, intensity, duration, climate, preparedness, response, demand, backup, network and repair controls
STRUCTURAL	**Geography + allocation**	CBS municipality/buurt context; audited NWB/GTFS/lifeline connectivity for Mobility and road-logistics Supply
CONTEXT	**Observed / official overlays**	KNMI, NDW, RWS, RIVM and utility geometry—timestamped and visible, but non-causal unless explicitly bound
UNKNOWN	**Unsupported or missing evidence**	NoData remains Unknown. It is never silently converted into zero impact, open roads or operational services.

Model contract: README “Data and provenance” + packages/sim-core

SOURCES

A Dutch public-data ecosystem

Official and public datasets are grouped by role, provenance and valid time.

PLACE + PEOPLE

CBS / PDOK
BAG / BRT TOP10NL

MOBILITY

NWB / ProRail
OVapi · NDOV / NDW / FIS

WEATHER + HAZARDS

KNMI / RWS Waterinfo / LIWO
AHN4 / RIVM / Risicokaart / RVO

ESSENTIAL SERVICES

Liander / Stedin / Netbeheer NL
HWH · DAMO · GWSW / RDI / ENTSOG

FACILITIES

KVK public services
RIVM acute care / DUO education

63,567	WATER ASSETS
32,270	ANTENNA SITES
12,883	EDUCATION LOCATIONS
79	ACUTE-CARE LOCATIONS

Every refresh preserves source URL, edition, licence, retrieval time and caveats.

Checked-in dataset counts from the current repository snapshot

EVIDENCE

Observed is not the same as simulated

Each layer tells the user what it can—and cannot—support.

OFFICIAL CONTEXT

Boundaries, assets, provider areas and registers

Where things are

OBSERVED CONTEXT

Radar, stations, incidents and current measurements

What was observed

SCENARIO STATE

Synthetic hazard, availability, demand and recovery

What the model explores

UNKNOWN / NODATA

Missing coverage, horizon, permission or semantics

What must not be inferred

NO DATA ≠ ZERO HAZARD ≠ OPEN ROUTE ≠ HEALTHY SERVICE

WHY THIS STATE?

Explainability is part of the engine

Every municipality frame returns its own drivers, stage deltas and counterfactuals.

SERVICE AVAILABILITY TRACE

BASELINE RESILIENCE	starting state
DIRECT HAZARD	exact stage change
RESTORATION	repair progress
UPSTREAM DEPENDENCY	power / mobility effect
INTERVENTION	selected policy effect
NETWORK TRANSFER	supported route effect

DRIVERS

Which inputs and assumptions were active?

CONTRIBUTIONS

Which stage increased or reduced availability?

DEPENDENCY CHAIN

Which upstream service caused the constraint?

COUNTERFACTUALS

What changes without damage, delay or dependency?

Generated by deterministic reruns—not post-hoc AI prose.

SCENARIO LAB

Ten starting points, many controlled variants

Presets provide a reproducible baseline; operators can then test assumptions.

WEATHER + WATER

- SEVERE WINDSTORM
- URBAN CLOUDBURST
- RIVER HIGH WATER
- COASTAL SURGE
- HEATWAVE + DROUGHT
- WINTER STORM

SYSTEM + GEOPHYSICAL

- CYBERATTACK
- GRONINGEN EARTHQUAKE
- WILDFIRE + SMOKE
- COMPOUND CASCADE

CONTROL LEVERS

season · KNMI'23 path + horizon · intensity · duration · response delay · infrastructure preparedness

day type · local start time · demand · backup · repair · network capacity · household preparedness · reference date

SAFETY GUARDS

Designed to fail safely

Unsupported local precision is withheld instead of rendered as confident detail.

245

OF 342 MUNICIPALITIES AUDITED

as of 3 September 2026

1 **AUDIT**
Scenario matrix, numerical invariants, spatial plausibility, dependencies and T+72 recovery

2 **COMPILE**
Correction-required or blocked findings become deterministic generated guards

3 **WITHHOLD**
Unsupported hazard pathways and downstream claims become Unknown / neutral

4 **BLOCK BUILD**
A missing or stale generated guard prevents production output

Audit prose never becomes a causal model automatically.

SCOPE

Powerful for exploration—careful about claims

The prototype is explicit about what is modeled, observed, assumed and unresolved.

WHAT IT DOES WELL

- Compare assumptions and response policies
- Reveal time-dependent service cascades
- Keep national and local views coherent
- Expose drivers and counterfactuals
- Preserve provenance and uncertainty

WHAT IT DOES NOT CLAIM

- A calibrated digital twin or forecast
- Live utility outage or capacity telemetry
- Physical power / water / telecom topology
- Observed affected-person headcounts
- Real-world units for generalized route capacity

The honest limitation is a feature: uncertainty stays visible.

LIVE DEMO

A four-minute walkthrough

Tell one causal story instead of showing every control.

1 **START NATIONAL**
Choose Compound cascade; keep the affected-region overlay visible.

2 **MOVE THROUGH TIME**
Scrub from T+0 to T+24 and call out the changing headline metrics.

3 **FOLLOW A DEPENDENCY**
Switch Power → Telecom, or Mobility → Supply, to reveal the cascade.

4 **ZOOM TO EINDHOVEN**
Open municipality and buurt detail; compare demand, delivery and reserve.

5 **CHANGE ONE LEVER**
Increase backup or reduce response delay—keep everything else constant.

6 **EXPLAIN THE RESULT**
Open “Why this state?” and finish with the exact causal trace.

Presenter rule: change one assumption, then show one downstream effect.

ROADMAP

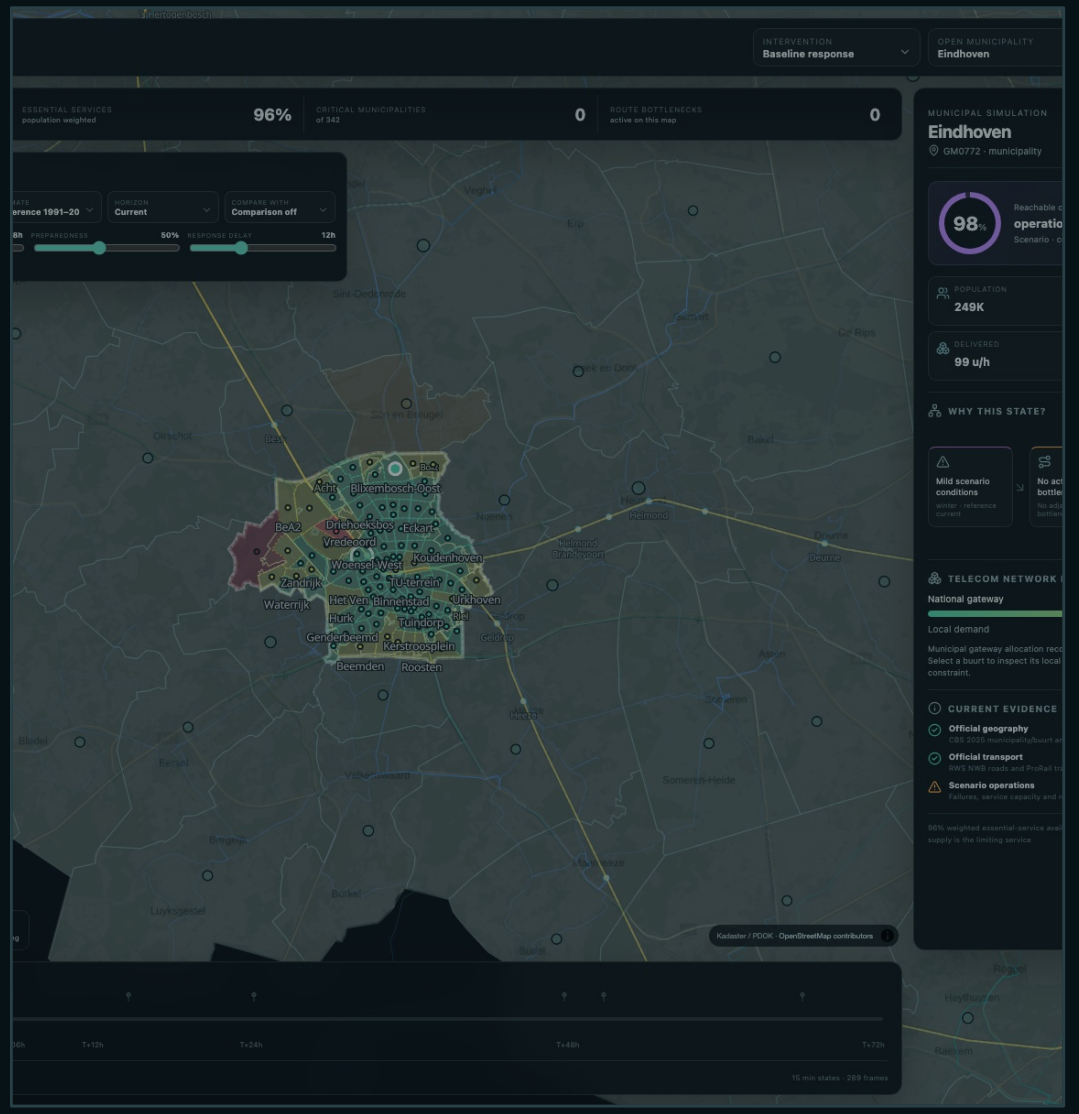
From exploration prototype to calibrated exercises

The next gains come from stronger event binding, topology and validation—not more visual layers.

01	EVENT-BOUND WEATHER	Bind checksum-versioned forecast cycles; preserve the T+60–T+72 coverage gap.
02	HYDRAULIC CALIBRATION	Select compatible LIWO / breach scenarios with depth, arrival and recession.
03	SERVICE-SPECIFIC NETWORKS	Add authorized topology, capacity, switching and restoration semantics.
04	FACILITIES + RESPONSE	Calibrate shelter, acute-care, staffing, generators, fuel and dispatch resources.
05	VALIDATION	Compare outputs with historical incidents and expert tabletop exercises.

CALIBRATION · GOVERNANCE · VALIDATION · EXERCISE DESIGN

See the cascade.
Question the assumptions.
Improve the response.



Questions?

GitHub — github.com/Rik-Stokmans/DisasterSimulator

Preview — rikstokmans.com/warroom/projects/disasterSimulator